⊛ Telescoping Series

Example:

$$S = \frac{1}{1*2} + \frac{1}{2*3} + \frac{1}{3*4} + \frac{1}{4*5} + \cdots + \frac{1}{K(K+1)} + \cdots$$

Notice that

$$\frac{1}{K\cdot(K+1)} = \frac{(K+1)-K}{K(K+1)} = \frac{(K+1)}{K(K+1)} - \frac{K}{K(K+1)}$$

$$= \frac{1}{K} - \frac{1}{K+1}$$

So $\frac{1}{1*2} = \frac{1}{1} - \frac{1}{2}$

$\frac{1}{2*3} = \frac{1}{2} - \frac{1}{3}$

$\frac{1}{3*4} = \frac{1}{3} - \frac{1}{4}$

$\frac{1}{4*5} = \frac{1}{4} - \frac{1}{5}$

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$$

$$S_n = \frac{1}{1\times 2} + \frac{1}{2\times 3} + \cdots + \frac{1}{n(n+1)} = 1 - \frac{1}{n+1}$$

AS $n \to \infty$

$$S_n \to 1 - 0 = 1$$

$$\Rightarrow \quad S = 1$$

Example:

$$S = \sum_{k=1}^{\infty} \frac{1}{k^2+3k+2} = \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \cdots$$

$$\frac{1}{(k+1)(k+2)} = \frac{1}{k+1} - \frac{1}{k+2}$$

$$\frac{1}{6} = \frac{1}{2*3} = \frac{1}{2} - \frac{1}{3}$$

$$\frac{1}{12} = \frac{1}{3*4} = \frac{1}{3} - \frac{1}{4}$$

+

$$\frac{1}{20} = \frac{1}{4*5} = \frac{1}{4} - \frac{1}{5}$$

$$\vdots$$

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$$

$$\Rightarrow S_n = \frac{1}{2*3} + \frac{1}{3*4} + \cdots + \frac{1}{n(n+1)}$$

$$= \frac{1}{2} - \frac{1}{n+1} \to 0$$

Let $n \to \infty$

$$\Rightarrow S = \frac{1}{2}$$

Example

$$S = \frac{1}{1*4} + \frac{1}{4*7} + \frac{1}{7*10} + \frac{1}{10*13} + \cdots$$

$$+ \frac{1}{(3k-2)(3k+1)} + \cdots\cdots$$

Noting that

$$\frac{1}{(3k-2)(3k+1)} = \frac{(3k+1)-(3k-2)}{(3k-2)(3k+1)} \cdot \frac{1}{3}$$

$$= \frac{1}{3}\left[\frac{3k+1}{(3k-2)(3k+1)} - \frac{3k-2}{(3k-2)(3k+1)}\right]$$

$$= \frac{1}{3}\left[\frac{1}{3k-2} - \frac{1}{3k+1}\right]$$

$$\Rightarrow \frac{1}{1\cdot 4} = \frac{1}{3}\left[\frac{1}{1} - \frac{1}{4}\right]$$

$$\frac{1}{4\cdot 7} = \frac{1}{3}\left[\frac{1}{4} - \frac{1}{7}\right]$$

$$\frac{1}{7\cdot 10} = \frac{1}{3}\left[\frac{1}{7} - \frac{1}{10}\right]$$

$$\frac{1}{(3n-2)(3n+1)} = \frac{1}{3}\left[\frac{1}{3n-2} - \frac{1}{3n+1}\right]$$

$$S_n = \frac{1}{1.4} + \frac{1}{4.7} + \cdots + \frac{1}{(3n-2)(3n+1)}$$

$$= \frac{1}{3}\left[1 - \frac{1}{3n+1}\right]$$

$$\frac{1}{3n+1} \to 0$$

$$\Rightarrow S_n \to \frac{1}{3} \Rightarrow \boxed{S = \frac{1}{3}}$$